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Sprint 3: Churn Prediction Project

Project Option: Data Science Lifecycle

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Executive Summary

In this project, I used machine learning to predict which telecom customers are likely to leave the service. Customer churn is a common issue in the telecom world. Companies lose money when customers cancel, and it costs more to get new ones. If a company can predict who might leave early, they can offer deals or support to keep them. I worked with a real dataset that had information about customers, their accounts, and services. After cleaning the data, I trained and tested a few models: logistic regression, decision trees, and random forest. Random forest gave the best results.
To organize everything, I followed the Cookiecutter Data Science structure. I also created charts to show which features were most useful and how well the models performed. This kind of model could help telecom companies find customers at risk of leaving and act early to keep them, saving money and improving customer satisfaction.

Problem Statement & Business Value

Customer churn is a serious issue for telecom businesses. Losing customers means losing income and spending more to get new ones. It’s hard to know exactly who will leave without a good system in place. I chose this topic because churn prediction is a real and important problem. It connects business concerns with data science skills.
The plan is to create a machine learning model that can describe which clients might leave soon. This way, the company can act before it's too late. With better predictions, they can target at risk customers with offers or better service. On the technical side, this

project is a good example of turning raw data into useful predictions. On the business side, it helps customer success teams and supports smart decision making.

Methodology

Data Preparation

I used a public telecom dataset with customer info like gender, service type, charges, contract details, and if they left or stayed. I loaded the data using pandas and looked for missing values or strange entries. Some columns had things like No internet service, which I cleaned up to make analysis easier. There were no missing values or duplicate rows so I moved on to formatting and encoding.

I used one hot encoding to turn text data into numbers so that machine learning models could use them. Then, I scaled numerical columns like "MonthlyCharges" using standard scaling. I split the data into training (80%) and testing (20%) sets.

Feature Engineering

Based on both the literature and personal exploration, I selected features that were most likely to influence churn. I focused on variables like contract type, payment method, tenure, monthly charges, and whether the customer had tech support.

I also checked correlations and removed redundant features to reduce noise. While I kept feature engineering simple in this version, I wanted to make sure I started with clean, meaningful inputs before moving on to modeling.

Modeling Approaches

I started with logistic regression as a baseline because it is interpretable and simple. After that, I tried a decision tree to see how a rule based model would perform. Finally, I trained a random forest classifier, which combines multiple trees and typically provides stronger performance. I used scikit learn for all models and made sure to split training and evaluation carefully. I also checked accuracy, precision, recall, and F1-score for all models so I could compare them on multiple metrics.

Hyperparameter Tuning

For the decision tree and random forest, I used GridSearchCV to perform hyperparameter tuning. I tested diferent tree depths and numbers of estimators to find the best combinations. I also used cross validation during this process to avoid overfitting and make sure that my models would generalize well to unseen data.

Model Evaluation & Selection

Here is a comparison of the models I used:

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	79%	68%	52%	59%

Decision Tree	80%	71%	58%	64%
Random Forest	83%	75%	61%	67%

Random forest was the best overall. It had higher precision and F1-score, which is important in churn problems. These metrics help balance false alarms (false positives) and missed cases (false negatives). I also looked at confusion matrices and feature importance. The most important features were contract type, tech support, tenure, and charges.

One common error was predicting new customers as likely to churn, even when they stayed. This makes sense because new customers often have a higher chance of leaving.

Implementation Strategy & Business Value

If used in a real telecom company, this model could help the customer team focus on people who are likely to leave. For example, the model showed that customers with month-to-month contracts and no tech support are at high risk. The company could offer these people discounts or switch them to yearly contracts.

To make this work in the real world, the model would need to be added to a company system, like a CRM. My project already has code for cleaning data, training the model, and making predictions, so it could be automated. One challenge would be keeping the model updated as customer behavior changes. Also, staff would need training to use the model’s results.

This kind of system could save money by reducing how many people leave. Even a small improvement in churn could have a big financial impact.

Ethical Considerations & Limitations

A major ethical problem is bias. If the data has unfair patterns like one group leaving more often, the model could make unfair predictions. This could lead to some customers being treated worse than others. It's important to check the model regularly for fairness.

Also, this dataset comes from one company. That means it might not work well for another telecom company with different services or customers. I also didn’t have data like customer reviews or complaint history. That kind of information could help improve predictions.

Another risk is giving offers to people who were never going to leave, which wastes money. That’s why it’s important to use precision and F1-score when judging model quality.

I recommend doing regular audits and fairness checks if this model is used in real life. It should support decisions not replace human judgment.

Conclusion & Future Work

This project helped me learn how machine learning can solve business problems. I went through the full data science process from cleaning data to training and comparing models. The random forest model worked best and gave balanced predictions.

I learned how important it is to choose features carefully and tune model settings. I also saw the value of clear charts and metrics when explaining the results to others. In the future, I want to try more advanced models like XGBoost or LightGBM. I also want to use methods like RandomizedSearch for faster tuning. Adding more data, like support call logs or customer reviews, might improve results.

If I continue this project, I'll focus on fairness and interpretability so that the model is both useful and ethical. This work showed me that machine learning can give companies better ways to make smart decisions about customer retention.

GitHub Repo: https://github.com/Bshifer1/INST414_Churn_Prediction_Project

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